

Learning Predator–Prey Interactions from Real-World Animal Behavior via Inverse Reinforcement Learning

Abstract—Deadline: 15.09.2026

Requirements:

- IEEE template
- max. 8 pages
- completely anonymous
- 3 keywords (peer-reviewing)
- optional: video submission

ToDo's:

- Source validation
- Joint GitHub
- Video animation (synthetic & imitation, biological & imitation)

I. INTRODUCTION

Collective behavior is one of the most striking phenomena in nature, seen in bird flocks, fish schools, and the foraging trails of ants [1], [2], [3]. Although each individual responds only to local information, groups can move as coherent units, changing direction and shape without centralized control [1], [2]. It is well-known that this collective order arises through self-organization, emerging from local interactions among individuals rather than external coordination [4]. Understanding how such behavior emerges is important not only for biology but also for engineering, as the principles underlying natural collective motion have inspired decentralized control strategies for robotic swarms.

Early computational models showed that coordinated collective motion can emerge from simple local interaction rules. Reynolds' Boids model [2] reproduced flocking through separation, alignment, and cohesion, while Couzin et al. [1] described fish schools using concentric zones of repulsion, orientation, and attraction. These agent-based models are simple, interpretable, and remarkably successful, but they rely on assumed interaction rules rather than measured ones. The form of the interactions is specified a priori, and model parameters are adjusted until the simulated groups reproduce macroscopic properties such as polarization or group shape. Such agreement, however, does not uniquely validate the underlying behavioral rules, because different interaction mechanisms can generate similar collective patterns. Experimental observations further show that real animals do not interact according to fixed spatial zones: the influence of neighbors depends on both spatial configuration and behavioral context [5]. Extending rule-based models to capture these effects typically requires introducing additional hand-crafted rules for each newly observed behavior, making the models increasingly difficult to generalize. Data-driven approaches take the opposite strategy. Rather than prescribing interaction

rules, they infer individual decision-making directly from behavioral observations.

Imitation learning (IL) provides a natural framework for learning collective behaviour from demonstrations and is commonly divided into behavioural cloning and inverse reinforcement learning (IRL) [6]. Behavioural cloning directly predicts actions from observations. For example, Heras et al. [5] use an attention-based interaction network to predict zebrafish turning decisions from a weighted aggregation of neighbouring fish, achieving an accuracy of 85%. Their model further shows that a fish attends to roughly 17 neighbours, although its behaviour is largely determined by about 5, and that this influential subset changes on a sub-second timescale. However, because behavioural cloning is trained only on states visited by the demonstrator, prediction errors accumulate during autonomous rollouts [7]. IRL instead infers the reward function that best explains the observed behaviour [8]. Schafer et al. [9] apply Bayesian IRL to tracked fish trajectories, obtaining a compact behavioural model at the cost of repeatedly solving an inner reinforcement learning problem. Adversarial imitation learning removes this requirement by training a discriminator to distinguish generated behaviour from demonstrations and using its output as the reward signal [10]. Rather than matching individual actions, it aligns the overall behavioural distribution, making it well suited to collective systems where population-level statistics are often the primary quantity of interest. Building on this idea, Yapei Wu et al. [11] combine adversarial imitation learning with an attention-based interaction network to reproduce milling, schooling, and swarming from Couzin-model demonstrations before transferring the learned policy to a swarm of 20 wheeled robots. Yu et al. [12] jointly learn a shared reward function and policy from Vicsek-model demonstrations, while Yifan Wu et al. [13] learn directly from raw fish videos using a masked video autoencoder, avoiding explicit trajectory reconstruction. Despite these methodological differences, all three approaches assume homogeneous swarms and control every individual with a single shared policy.

Predator–prey systems break this assumption of homogeneity. Predators and prey pursue opposing objectives, and the collective dynamics emerge from the continuous interaction between pursuit and evasion rather than from interactions within either group alone. Li et al. [14] showed that this coupling is sufficient to generate complex collective behaviour: agents trained by reinforcement learning with only a survival reward, without explicit alignment or cohesion terms, spontaneously exhibited flocking, predator confusion,

and marginal predation. However, these behaviours emerged from hand-designed reward functions rather than demonstrations of real animals. By contrast, existing imitation learning methods learn directly from demonstrations but assume a single shared policy for all agents. Even when multiple roles are present, as in [13], they arise from different reward weights within a common policy instead of being learned as distinct, interacting behaviours. Learning separate predator and prey policies directly from demonstrations therefore remains an open problem.

To address this problem, we propose an adversarial imitation learning framework that models predators and prey as separate agents, each equipped with its own policy and discriminator. The two policies are trained jointly through alternating optimization with an evolutionary strategy, allowing them to adapt to one another while remaining stable during learning. We first validate the framework on synthetic predator–prey demonstrations generated with the Couzin model and then apply it to real biological predator–prey recordings [15]. Our main contributions are threefold: (i) a role-asymmetric adversarial imitation learning framework with separate policies and discriminators for predators and prey; (ii) the application of alternating co-training to predator–prey learning with conflicting role-specific objectives; and (iii) validation on both synthetic and real datasets, demonstrating that the learned policies reproduce key features of predator–prey interactions.

[Replace the last clause with your concrete quantitative result once available.]

II. MULTI-SPECIES BEHAVIORAL MODELING FRAMEWORK

The proposed framework consists of two role-specific policies that interact in a shared two-dimensional environment. A dedicated policy models the distinct predator behavior, while a single parameter-shared policy controls all prey individuals. The policies generate rollouts that are transformed into the same state representation as the corresponding expert demonstrations. Separate transition encoders and discriminators evaluate whether predator and prey transitions resemble their respective expert behavior. The resulting discriminator feedback is used to optimize both policies alternately, allowing them to co-adapt through their interactions in the shared environment.

A. Role-specific Policy Architecture

The predator and prey policies are denoted by $\pi_{P,\phi,\varphi}$ and $\pi_{Y,\phi,\varphi}$, respectively. Both policies follow the modular architecture introduced by Heras et al. [5] and later adopted by Yapei Wu et al. [11], consisting of a pairwise interaction network ϕ (PIN) and an attention network φ (AN). This decomposition separates neighbor-specific responses from their aggregation, preserving interpretability while keeping the policy scalable with the number of neighbors. Both networks use three fully connected hidden layers with 128, 64, and 32 units and ReLU activations, moderately extending

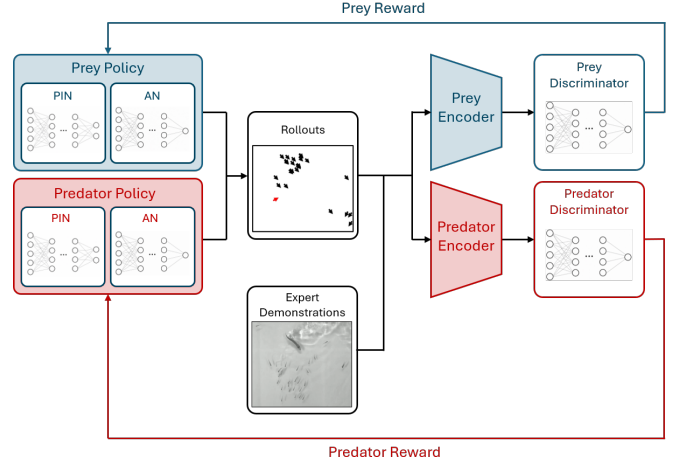


Fig. 1. GAIL architecture for multi-species modeling.

the architecture of [11] for the more complex real predator–prey trajectories. For each focal–neighbor pair, the PIN processes the relative position and velocity features introduced in Section III-D. The prey input additionally contains a binary predator flag, enabling the policy to distinguish the predator from conspecific neighbors. The PIN predicts the mean μ_i and standard deviation σ_i of a Gaussian distribution from which a neighbor-specific turning response is sampled:

$$a_i \sim \mathcal{N}(\mu_i, \sigma_i). \quad (1)$$

This stochastic formulation introduces behavioral variability instead of producing identical reactions under identical local conditions. The resulting sample diversity further helps mitigate the risk of mode collapse.

In parallel, the AN assigns an attention weight ω_i to each focal–neighbor interaction, representing the relative influence of neighbor i on the focal individual’s decision. The final policy output is obtained by aggregating the neighbor-specific responses through a normalized weighted sum:

$$a = \sum_{i \in \mathcal{I}} a_i \frac{\omega_i}{\sum_{j \in \mathcal{I}} \omega_j}, \quad (2)$$

The same aggregation principle is applied to both roles. For the prey policy, the predator is not handled through a separate aggregation rule. Instead, its role is encoded by the predator flag, allowing the policy to learn role-specific responses directly from the demonstrations. The aggregated output defines the per-step heading change of the focal individual.

B. Latent Transition Encoding and Discrimination

To provide the discriminator with compact information about temporal state changes, separate transition encoders $E_{P,\psi}$ and $E_{Y,\psi}$ are used for predator and prey trajectories, respectively. Directly processing complete state sequences would result in a high-dimensional discriminator input that grows with the number of frames, agents, neighbors, and features. This design was inspired by Yifan Wu et al. [13],

who modeled collective behavior using latent transitions between consecutive video frames.

For each agent and time step, neighbor-level information is first aggregated and mapped to a latent agent state z_t . The transition between two consecutive states is represented as

$$e_t = [z_t, \Delta z_t], \quad \Delta z_t = z_{t+1} - z_t, \quad (3)$$

thereby preserving both the current latent state and its temporal change.

The encoders are pretrained separately on the corresponding expert trajectories using Variance-Invariance-Covariance Regularization (VICReg) [16]. The self-supervised objective encourages augmented views of the same trajectory sequence to obtain similar representations while preventing representational collapse. After pretraining, the projectors are discarded and the encoder parameters are frozen. Consequently, both expert and policy-generated trajectories are encoded within the same fixed latent space during adversarial training.

The resulting transition embeddings are evaluated by separate role-specific discriminator $D_{P,\beta}$ and $D_{Y,\beta}$. Each discriminator operates on the embeddings of its corresponding role and assigns higher Wasserstein scores to transition patterns that increasingly resemble the expert demonstrations. A gradient penalty regularizes the discriminator and encourages stable adversarial training. The resulting score difference between expert and policy-generated transitions provides the role-specific feedback used to optimize the predator and prey policies.

C. Co-Evolutionary Training

Before adversarial training, the predator and prey policies are initialized through behavioral cloning (BC) [17]. For this purpose, the expected value of the predicted action distribution is trained to match the expert heading changes using a mean squared error objective. This provides both policies with an initial behavioral direction instead of starting the adversarial optimization from randomly initialized parameters.

During each training generation, exponential moving average (EMA) copies of the predator and prey policies are executed in the shared two-dimensional environment. The EMA policies smooth short-term parameter fluctuations and thereby provide more stable rollout distributions for discriminator training [18]. The corresponding expert demonstrations are denoted by $\tau_{E,P}$ for the predator and $\tau_{E,Y}$ for the prey. The environment is initialized from states sampled from the expert demonstrations, ensuring that rollouts start from behaviorally relevant configurations. Policy actions define the per-step heading changes, while positions are updated using a constant movement speed of 10 pixels per step, reducing the policy action space to steering decisions only. The resulting rollouts are transformed into the same state and transition representation as the expert trajectories, allowing the discriminators to compare both sources within the same latent space.

The role-specific discriminators are updated on the encoded expert and policy-generated transition features using

Root Mean Square Propagation [19]. The discriminator-update ratio controls how many discriminator optimization steps are performed before each policy update, helping to balance the learning progress of discriminators and policies and preventing either side from dominating the adversarial training. Multiple discriminator updates are performed before each policy update to mitigate training imbalances and maintain informative reward signals. The predator and prey policies are subsequently optimized using OpenAI Evolution Strategies (ES) [20]. For each policy module, mirrored Gaussian perturbations $\theta + \sigma \epsilon_i$ and $\theta - \sigma \epsilon_i$ are evaluated from identical initial states. The difference between their discriminator-based returns determines the direction of the parameter update. Mirrored sampling and rank normalization reduce estimator variance and limit the influence of extreme fitness values [11].

The predator and prey policy modules are optimized alternately. Decomposing the networks into separate role-specific optimization steps reduces the complexity of each individual update. At the same time, both policies continue to co-adapt, since each update changes the behavior encountered by the opposing role in subsequent rollouts. Training is further stabilized through ES update clipping, discriminator input noise during the early exploration phase, and a gradual decay γ of the learning rates λ_P, λ_Y and exploration factors σ_P, σ_Y as optimization progresses.

Algorithm 1 GAIL training with OpenAI-ES

Input: Policies for predator $\pi_{P,\phi,\varphi}$ and prey $\pi_{Y,\phi,\varphi}$,
Discriminators $D_{P,\beta}$ and $D_{Y,\beta}$,
Encoders $E_{P,\psi}$ and $E_{Y,\psi}$,
Expert trajectories $\tau_{E,P}$ and $\tau_{E,Y}$

Output: Best $\pi_{P,\phi,\varphi}^*$ and $\pi_{Y,\phi,\varphi}^*$

- 1: Pretrain $\pi_{P,\phi,\varphi}$ and $\pi_{Y,\phi,\varphi}$ with BC
 - 2: Train and freeze $E_{P,\psi}$ and $E_{Y,\psi}$
 - 3: Initialize EMA policies $\bar{\pi}_{P,\phi,\varphi}$ and $\bar{\pi}_{Y,\phi,\varphi}$
 - 4: **for** $i = 0, 1, 2, \dots, M$ generations **do**
 - 5: Execute $\bar{\pi}_{P,\phi,\varphi}$ and $\bar{\pi}_{Y,\phi,\varphi}$ to collect rollouts
 - 6: **for** $j = 1, \dots, J$ $D_{P,\beta}$ -update-ratio **do**
 - 7: Update predator discriminator $D_{P,\beta}$
 - 8: **end for**
 - 9: **for** $j = 1, \dots, J$ $D_{Y,\beta}$ -update-ratio **do**
 - 10: Update prey discriminator $D_{Y,\beta}$
 - 11: **end for**
 - 12: optimize with ES: PIN $\pi_{P,\phi}$ and AN $\pi_{P,\varphi}$
 PIN $\pi_{Y,\phi}$ and AN $\pi_{Y,\varphi}$
 - 13: update EMA: $\bar{\pi}_{P,\phi,\varphi} \leftarrow \text{EMA}(\pi_{P,\phi,\varphi})$
 $\bar{\pi}_{Y,\phi,\varphi} \leftarrow \text{EMA}(\pi_{Y,\phi,\varphi})$
 - 14: apply decay γ to: learning rates λ_P, λ_Y
 exploration factors σ_P, σ_Y
 - 15: **end for**
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III. FROM SYNTHETIC VERIFICATION TO BIOLOGICAL IMITATION

Before applying the framework to biological observations, its components were verified under increasingly com-

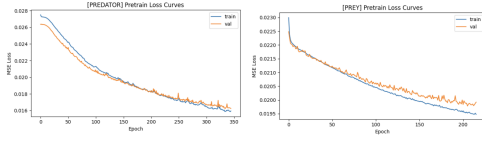


Fig. 2. Behavioral cloning pre-training loss curves for predator and prey.

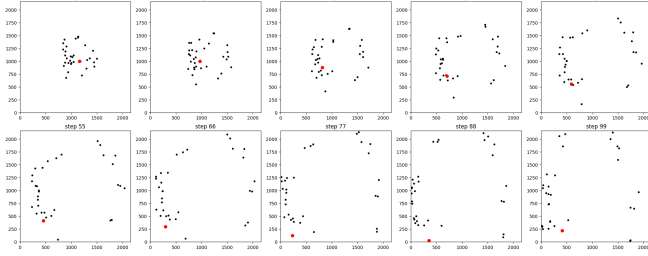


Fig. 3. Behavioral cloning roll outs.

plex conditions. First, behavioral cloning was evaluated to check whether role-specific variable-neighborhood state representation could be learned from expert demonstrations. The complete co-evolutionary training procedure was then evaluated on synthetic predator-prey demonstrations with predefined interaction dynamics. After verifying the role-specific architecture and co-evolutionary training procedure, the same modeling framework was transferred to trajectories reconstructed from real predator-prey recordings.

A. Behavior Cloning

As a preliminary step, the behavior cloning procedure was verified independently of the adversarial training loop. Both predator and prey policies show a smooth MSE decay in training and validation loss, confirming that the representation is learnable and providing a stable initialization for the co-evolutionary training stages applied to the biological-data experiments below. While behavioral cloning does produce plausible per-step actions, the roll outs demonstrate that the resulting policies do not sustain cohesive group structure or directed pursuit, motivating the progression to GAIL.

B. Couzin-Based Synthetic Verification

To generate the synthetic expert demonstrations, we adapted the predator-prey simulation provided by Haeri [21], which extends the original Couzin model [1] by introducing a predator. Prey motion follows the local repulsion, orientation, and attraction rules of the Couzin model. Simulations were conducted in a 50×50 two-dimensional arena for 500 steps, using a constant agent speed of 5, a step size of 0.5, and a maximum per-step turn of 0.25.

In the predator-prey simulation, a single predator follows a deterministic pursuit rule targeting the nearest visible prey [21]:

$$i_t^* = \arg \min_{i \in I} \|\mathbf{x}_{i,t} - \mathbf{x}_{P,t}\|, \quad (4)$$

where I denotes the set of prey visible to the predator. The corresponding desired pursuit direction is given by

$$\mathbf{d}_{P,t} = \frac{\mathbf{x}_{i_t^*,t} - \mathbf{x}_{P,t}}{\|\mathbf{x}_{i_t^*,t} - \mathbf{x}_{P,t}\|}. \quad (5)$$

The predator turns toward this direction subject to the same bounded-turn dynamics as the remaining agents and subsequently moves at constant speed. Prey retain the Couzin interaction rules, augmented by predator avoidance within a predefined threat radius. The resulting trajectories provide clean and gap-free demonstrations with predefined interaction dynamics, avoiding uncertainties introduced by the biological tracking pipeline. After verifying the framework on both homogeneous collective motion and synthetic predator-prey interactions, the same architecture was applied to trajectories reconstructed from biological recordings.

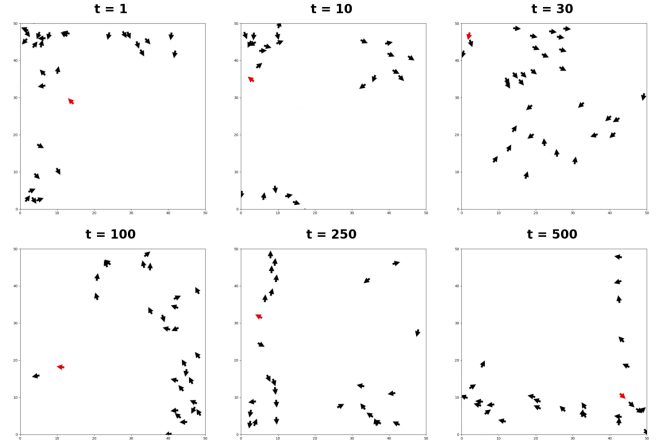


Fig. 4. Couzin-simulated Predator-Prey trajectories [1], [22]

C. Biological Recordings and Trajectory Reconstruction

The recordings capture predator-prey interactions between groups of sunbleaks (*Leucaspius delineatus*) and a northern pike (*Esox lucius*) in a controlled laboratory aquarium [15]. Trials were conducted in a featureless 1.5×1.5 m tank, where groups of prey fish were introduced before the predator was released following an initial acclimatization phase. Visual shielding, vibration damping, and controlled environmental conditions were used to minimize external disturbances. Behavior was recorded from above using a four-camera array.

After selecting recordings in which the predator was present, the dataset used in this study consisted of 35 videos with a resolution of 2160×2160 pixels at 30 fps and a total duration of 16 h 42 min 49 s. The trajectory dataset was restricted to configurations containing one predator and 32 prey individuals. Active pursuit behavior was strongly under-represented: 32 attack events were observed, corresponding to 18.9 s of footage and approximately 0.12% of the total recording time.

Individual trajectories were extracted using a detection-and-tracking pipeline based on a fine-tuned YOLOv11 model [23] and DeepSORT [24]. YOLOv11 detected prey individuals and the predator head, while DeepSORT associated these detections across consecutive frames to preserve



Fig. 5. Example frame from the experimental predator–prey recordings.

individual identities. Fish positions were represented by the center points of their corresponding bounding boxes. Predator detection was highly stable, with a mean absolute count error of 0.03 fish per frame. Prey detection was more strongly affected by occlusions, reflections, shadows, and dense schooling behavior, resulting in a mean absolute count error of 3.11 fish per frame.

The reconstructed trajectories were filtered to retain only continuous sequences containing exactly one predator and 32 prey with consistent identities. Frames without a predator detection were discarded, and implausible speed jumps caused by detection errors or identity switches were removed. The remaining trajectories were visually inspected and divided into fixed-length temporal windows for subsequent state construction and imitation learning. Using a window length of ten frames, this procedure yielded 793 valid trajectory windows.

D. State Engineering and Tensor Construction

Both synthetic and reconstructed biological trajectories were transformed into the same focal-centered interaction representation. Each focal–neighbor pair is described by relative position and relative velocity in the focal coordinate frame [5], [11]. The prey representation additionally includes a binary predator flag to distinguish the predator from conspecifics. Consecutive states are stacked into fixed-length trajectory windows.

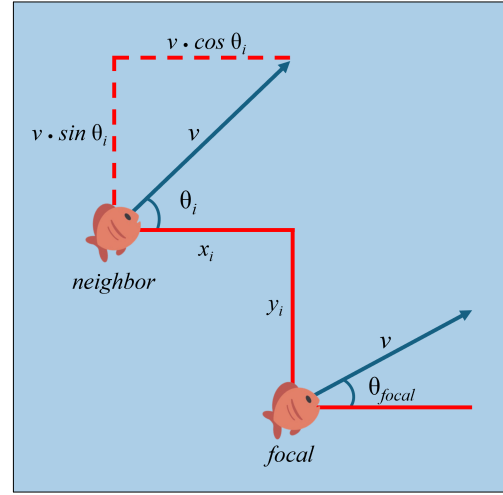


Fig. 6. State definition for focal individual [11].

Synthetic and biological trajectories are thus mapped to the same role-specific model interface. Consequently, the policy, transition encoder, and discriminator architectures remain unchanged across the verification and biological imitation stages. Differences between both settings arise from the underlying expert demonstrations and environment-specific movement parameters rather than from changes to the general framework.

IV. EXPERIMENTS AND RESULTS

A. Couzin-Based Predator–Prey Verification

First, we show that the framework can imitate known predator–prey dynamics under controlled conditions.

- Brief analysis of predator and prey training, including convergence plots and a brief description of Sinkhorn metrics and MMD.
- Couzin Expert vs. Couzin Imitation.
- Evaluation using the same predator–prey metrics that are later applied to the biological data.

B. Biological Predator–Prey Imitation

Next, the same evaluation is transferred to the real fish trajectories.

- Brief analysis of training and convergence plots.
- Biological Expert vs. Biological Imitation.
- Same metrics and the same plot structure.
- Direct assessment of how imitation quality changes from the controlled synthetic setting to the real biological system.

C. Generalization Across Group Sizes

We then evaluate whether the learned behavior remains consistent across different group sizes.

- 16 prey vs. 32 prey.
- Ideally without modifying or retraining the underlying policy.
- Evaluation using predator–prey metrics.
- Couzin and biological settings are again considered in parallel.

D. Analysis of Learned Interaction Mechanisms

Finally, we investigate how and why the observed behavioral patterns emerge from the learned policies.

- Pairwise Interaction Network.
- Attention Network.
- Separate analysis of Predator $\rightarrow$ Prey and Prey $\rightarrow$ Predator interactions.
- Direct comparison between Couzin-trained and biologically trained policies.

The modular structure of the policy networks enables a more detailed interpretation of the learned decision-making process [5], [11]. The PIN maps visualize how the position of an individual neighbor influences the focal individual's turning response, represented by predicted heading changes within $[-180^\circ, 180^\circ]$. The AN maps show how relative neighbor positions are weighted during action aggregation, with attention values ranging from low importance (0) to high importance (1).

For the prey policy, nearby conspecifics induce intuitive steering responses. Neighbors located to the left or right of the focal individual generally elicit corresponding left- or right-turn tendencies. Neighbors positioned directly behind the focal individual receive comparatively high attention but induce only minor steering changes (Fig. 8). More extreme PIN responses occur in far-field regions. However, these areas are typically assigned low attention by the AN and therefore contribute less strongly to the final aggregated action.

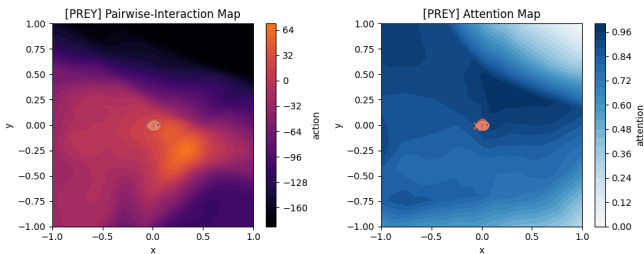


Fig. 7. Prey PIN and AN maps over relative neighbor positions.

The predator maps exhibit a clearer and less noisy attention structure (Fig. 7). Prey located in front of the predator receive the highest attention, whereas individuals positioned behind it are down-weighted. This suggests that the predator policy learned a forward-oriented pursuit tendency. Extreme PIN outputs remain visible in some regions, but these responses are primarily associated with low attention weights and therefore have limited influence on the final action.

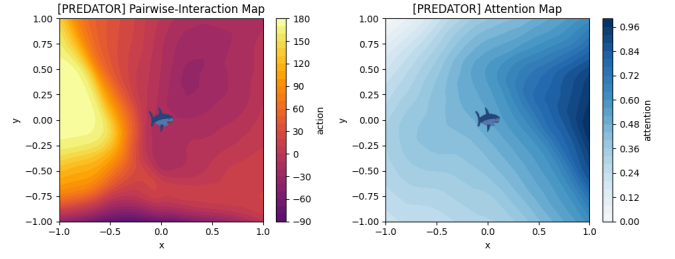


Fig. 8. Predator PIN and AN maps over relative neighbor positions.

V. DISCUSSION

- Controlled verification vs. biological imitation (main finding first).
- Critical reflection of training: What was learned and what was not learned?
- Generalization across group sizes.
- Pairwise Interaction Network and Attention Network analysis.
- Limitations.

The synthetic verification provides an important reference for interpreting the biological imitation results. Since synthetic and biological trajectories use the same state representation and model architecture, differences in imitation quality mainly reflect the underlying expert demonstrations and environment-specific dynamics rather than structural changes to the framework. Stronger performance on the Couzin-generated trajectories would indicate that the framework can recover controlled predator-prey dynamics, while remaining gaps on biological data would point toward the additional complexity and imperfections of real observations.

The biological setting introduces several limitations. Active pursuit behavior is strongly underrepresented, with only 32 observed attack events accounting for approximately 0.12% of the total recording duration. In addition, occlusions, reflections, and dense schooling reduce trajectory completeness and restrict the available temporal context. The constant-speed simulation further simplifies the observed dynamics by restricting the policy action space to steering decisions only.

VI. CONCLUSIONS

As it's not necessary, drop. Main conclusions are summarized in the abstract.

A conclusion section is not required. Although a conclusion may review the main points of the paper, do not replicate the abstract as the conclusion. A conclusion might elaborate on the importance of the work or suggest applications and extensions.

APPENDIX

Appendices should appear before the acknowledgment.

ACKNOWLEDGMENT

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